Lab 4 ticket

Inverses, Rank, and Determinants — Friday, November 6

Lab 4 — Inverses, Rank, and Determinants · Fri Nov 6 · ~25 min · Laptop · Handwrite · Initials

Name: _____ **Language (circle):** Octave / Python

New syntax

I needed to…	Command / pattern
Inverse	
Determinant	
Rank / null space	
Check $A^{-1}A$	

$A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$. Quiz 9 = Lecture 22 (**not** this lab).

C4.0. $A^{-1} =$ _____ $A^{-1}A \approx$ _____ **Expected:** I.

C4.1. $\det A =$ _____ Formula matches software? _____

C4.2. `inv(B)` — what happens? _____

C4.3. $\text{rank}(B) =$ _____ Nonzero $n \in N(B)$: _____ Nullity = _____ $(n - \text{rank})$

C4.4. $\det A =$ _____ $\det B =$ _____
Meaning ($\neq 0$ vs $= 0$): _____
Optional triangular det: _____

Stuck on: _____

Check-in initials: _____